

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

TRIGONOMETRIC FUNCTIONS

DIBY (DO IT BY YOURSELF)

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DIBY-01

If $\sin A \tan A = \cos^2 A$, then $\cos^3 A + \cos^2 A$ is equal to

- (A) 1 (B) 2 (C) 4 (D) None of these

Ans: (A)

DIBY-02

If $\operatorname{cosec} A + \cot A = \frac{11}{2}$, then $\tan A$ is

- (A) $\frac{21}{22}$ (B) $\frac{15}{16}$ (C) $\frac{44}{117}$ (D) $\frac{117}{43}$

Ans: (C)

DIBY-03

If $\tan \theta = \frac{-4}{3}$, then $\sin \theta$ is

- (A) $\frac{-4}{5}$ but not $\frac{4}{5}$ (B) $\frac{-4}{5}$ or $\frac{4}{5}$ (C) $\frac{4}{5}$ but not $-\frac{4}{5}$ (D) None of these

Ans: (B)

DIBY-04

If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, then prove that $m^2 - n^2 = 4\sin\theta \tan\theta$

DIBY-05

Prove that $\frac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \frac{1 + \sin A}{\cos A}$

DIBY-06

Prove that: $\sin^8 A - \cos^8 A = (\sin^2 A - \cos^2 A)(1 - 2\sin^2 A \cos^2 A)$

DIBY-07

$$(1 + \tan\alpha \tan\beta)^2 + (\tan\alpha - \tan\beta)^2 =$$

(A) $\tan^2\alpha \tan^2\beta$

(B) $\sec^2\alpha \sec^2\beta$

(C) $\tan^2\alpha \cot^2\beta$

(D) $\sec^2\alpha \cos^2\beta$

Ans: (B)

DIBY-08

Prove that: $\frac{\sin A}{1+\cos A} + \frac{1+\cos A}{\sin A} = 2\operatorname{cosec} A$

DIBY-09

Prove that: $\sqrt{\frac{1-\sin A}{1+\sin A}} = \sec A - \tan A$

DIBY-10

Prove that: $\frac{\operatorname{cosec} A}{\operatorname{cosec} A - 1} + \frac{\operatorname{cosec} A}{\operatorname{cosec} A + 1} = 2\sec^2 A$

DIBY-11

Prove that: $\frac{\operatorname{cosec} A}{\cot A + \tan A} = \cos A$

DIBY-12

Find value of following

- (i) $\sin^2 \frac{\pi}{6} + \cos^2 \frac{\pi}{3} - \tan^2 \frac{\pi}{4} =$
- (ii) $2\sin^2 \frac{\pi}{6} + \operatorname{cosec}^2 \frac{7\pi}{6} \cos^2 \frac{\pi}{3} =$
- (iii) $\cot^2 \frac{\pi}{6} + \operatorname{cosec} \frac{5\pi}{6} + 3\tan^2 \frac{\pi}{6} =$
- (iv) $2\sin^2 \frac{3\pi}{4} + 2\cos^2 \frac{\pi}{4} + 2\sec^2 \frac{\pi}{3} =$

Ans. (i) $\frac{-1}{2}$, (ii) $\frac{3}{2}$, (iii) 6, (iv) 10

DIBY-13

$\frac{\tan(90^\circ - \theta) \sec(180^\circ - \theta) \sin(-\theta)}{\sin(180^\circ + \theta) \cos(360^\circ + \theta) \operatorname{cosec}(90^\circ - \theta)}$ is equals

- (A) 0
- (B) 1
- (C) -1
- (D) 2

Ans. (D)

DIBY-14

$$\sec(270^\circ - A) \sec(90^\circ - A) - \tan(270^\circ - A) \tan(90^\circ + A)$$

Ans. -1

DIBY-15

$$\cot A + \tan(180^\circ + A) + \tan(90^\circ + A) + \tan(360^\circ - A)$$

Ans. 0

DIBY-16

$$\frac{\tan(180^\circ - \theta) \cos(180^\circ - \theta) \tan(90^\circ - \theta)}{\sin(90^\circ + \theta) \cot(90^\circ - \theta) \tan(90^\circ + \theta)}$$

Ans. -1

DIBY-17

If $\tan A = 2$ and $A \in \left(\pi, \frac{3\pi}{2}\right)$ then the expression $\frac{\cos A}{\sin^3 A + \cos^3 A}$ is equal to

Ans. 5/9

DIBY-18

$\sin^2 \frac{\pi}{18} + \sin^2 \frac{\pi}{9} + \sin^2 \frac{7\pi}{18} + \sin^2 \frac{4\pi}{9}$ equal

(A) 0

(B) 1

(C) -1

(D) 2

Ans. (D)

DIBY-19

Prove that

1. $\tan 720^\circ - \cos 270^\circ - \sin 150^\circ \cos 120^\circ = \frac{1}{4}$

2. $\sin 780^\circ \sin 480^\circ + \cos 120^\circ \sin 150^\circ = \frac{1}{2}$

DIBY-20

Prove that $\sec\left(\frac{3\pi}{2} - \theta\right) \sec\left(\theta - \frac{5\pi}{2}\right) + \tan\left(\frac{5\pi}{2} + \theta\right) \tan\left(\theta - \frac{3\pi}{2}\right) = -1.$

DIBY-21



If $\sin x = -\frac{3}{5}$ where $\pi < x < \frac{3\pi}{2}$, then $80(\tan^2 x - \cos x)$ is equal to

(A) 109

(B) 108

(C) 19

(D) 18

Ans. (A)

DIBY-22

1. $\sin(405^\circ)$ [Ans. $\frac{1}{\sqrt{2}}$]

2. $\sec(-1470^\circ)$ [Ans. $\frac{2}{\sqrt{3}}$]

3. $\tan(-300^\circ)$ [Ans. $\sqrt{3}$]

4. $\cot(585^\circ)$ [Ans. 1]

5. $\operatorname{cosec}(-750^\circ)$ [Ans. -2]

6. $\cos(-2220^\circ)$ [Ans. 1/2]

Prove that

$$1. \quad \frac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta} = \tan \theta$$

$$3. \quad \frac{\sin A + \sin 3A}{\cos A + \cos 3A} = \tan 2A$$

$$5. \quad \frac{\cos 2B + \cos 2A}{\cos 2B - \cos 2A} = \cot(A + B)\cot(A - B)$$

$$7. \quad \frac{\sin A + \sin 2A}{\cos A - \cos 2A} = \cot \frac{A}{2}$$

$$9. \quad \frac{\cos 2B - \cos 2A}{\sin 2B + \sin 2A} = \tan(A - B)$$

$$10. \quad \cos(A + B) + \sin(A - B) = 2\sin(45^\circ + A)\cos(45^\circ + B)$$

$$11. \quad \frac{\cos 3A - \cos A}{\sin 3A - \sin A} + \frac{\cos 2A - \cos 4A}{\sin 4A - \sin 2A} = \frac{\sin A}{\cos 2A \cos 3A}$$

$$2. \quad \frac{\cos 6\theta - \cos 4\theta}{\sin 6\theta + \sin 4\theta} = -\tan \theta$$

$$4. \quad \frac{\sin 7A - \sin A}{\sin 8A - \sin 2A} = \cos 4A \sec 5A$$

$$6. \quad \frac{\sin 2A + \sin 2B}{\sin 2A - \sin 2B} = \frac{\tan(A + B)}{\tan(A - B)}$$

$$8. \quad \frac{\sin 5A - \sin 3A}{\cos 3A + \cos 5A} = \tan A$$

DIBY-24

Find the value of expression $\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta}$, given $\theta = 7.5^\circ$.

- (A) $2 + \sqrt{3}$ (B) $2 - \sqrt{3}$ (C) 0 (D) 1

Ans. (B)

DIBY-25

If $\alpha = \frac{\pi}{19}$, find the value of $\frac{\sin 23\alpha - \sin 3\alpha}{\sin 16\alpha + \sin 4\alpha}$.

- (A) 1 (B) 0 (C) 2 (D) -1

Ans. (D)

DIBY-26

The expression $\frac{(1+\tan 37^\circ)(1+\tan 8^\circ)(1+\tan 15^\circ)(1+\tan 30^\circ)}{(1+\tan 2^\circ)(1+\tan 43^\circ)}$ reduces to

- (A) $2 \tan 45^\circ$ (B) 4 (C) 8 (D) $6 \cot 45^\circ$

Ans. (A)

DIBY-27

For $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, let $3\sin(\alpha + \beta) = 2\sin(\alpha - \beta)$ and a real number k be such that $\tan \alpha = k \tan \beta$. Then the value of k is equal to

- (A) 5 (B) $-2/3$ (C) -5 (D) $2/3$

Ans. (C)

DIBY-28

Let $\frac{\tan 50^\circ - \tan 40^\circ}{\tan 10^\circ} = k$, find $k \in N$.

Ans. 2

DIBY-29

If $x = \frac{\pi}{48}$, find the value of $\frac{\sin 7x + \sin 5x + \sin 9x + \sin 3x}{\cos 7x + \cos 5x + \cos 9x + \cos 3x}$.

Ans. $\sqrt{2} - 1$

DIBY-30

If $3\sin\alpha = 5\sin\beta \neq 0$, then the value of $\frac{\tan\left(\frac{\alpha+\beta}{2}\right)}{\tan\left(\frac{\alpha-\beta}{2}\right)}$ is:

Ans. 4

DIBY-31

the value of $\cos 24^\circ - \cos 12^\circ + \cos 48^\circ - \cos 84^\circ =$

Ans. $1/2$

DIBY-32



If $\cos\alpha \cos\beta = \frac{1}{2}$ and $\sin\alpha \sin\beta = \frac{1}{3}$

Column - I	Column - II
(A) $\cos \frac{\alpha+\beta}{2}$	(P) $\pm \frac{\sqrt{13}}{12}$
(B) $\cos \frac{\alpha-\beta}{2}$	(Q) $\frac{2}{3}$
(C) $\tan \left(\frac{\alpha+\beta}{2} \right)$	(R) $\pm \frac{3}{\sqrt{13}}$
(D) $\tan \left(\frac{\alpha-\beta}{2} \right)$	(S) $\pm \sqrt{\frac{131}{13}}$

Ans. A-R, B-P, C-Q, D-S

DIBY-33

Prove that $\frac{\sin \theta + \sin 2\theta}{1 + \cos \theta + \cos 2\theta} = \tan \theta$

DIBY-34

Prove that $\frac{1 - \cos A + \cos B - \cos(A+B)}{1 + \cos A - \cos B - \cos(A+B)} = \tan \frac{A}{2} \cot \frac{B}{2}$

DIBY-35



Value of : $\prod_{k=1}^{17} \sin \frac{k\pi}{18} = \frac{a}{2^{16}}$ then $a = \text{-----}$.

Ans. 9

DIBY-36

If $\frac{5\pi}{2} < x < 3\pi$, then the value of the expression $\frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}}$ is

(A) $-\cot \frac{x}{2}$

(B) $\cot \frac{x}{2}$

(C) $\tan \frac{x}{2}$

(D) $-\tan \frac{x}{2}$

Ans. (C)

DIBY-37

If $A = 340^\circ$ then $2\sin \frac{A}{2}$ is identical to

(A) $\sqrt{1 + \sin A} + \sqrt{1 - \sin A}$

(B) $-\sqrt{1 + \sin A} - \sqrt{1 - \sin A}$

(C) $\sqrt{1 + \sin A} - \sqrt{1 - \sin A}$

(D) $-\sqrt{1 + \sin A} + \sqrt{1 - \sin A}$

Ans. (D)

DIBY-38

Evaluate: $\cos^6 \frac{\pi}{16} + \cos^6 \frac{3\pi}{16} + \cos^6 \frac{5\pi}{16} + \cos^6 \frac{7\pi}{16}$

(A) $5/8$

(B) $5/4$

(C) $7/16$

(D) $5/32$

Ans. (B)

DIBY-39

If $\frac{x}{\cos \theta} = \frac{y}{\cos\left(\theta + \frac{2\pi}{3}\right)} = \frac{z}{\cos\left(\theta - \frac{2\pi}{3}\right)}$, then $x + y + z$ is equal to:

(A) -1

(B) 1

(C) 0

(D) 2

Ans. (C)

DIBY-40

Find the exact value of $\tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{7\pi}{16}$

Ans. 28

DIBY-41

Prove that $\cos 36^\circ \cos 72^\circ \cos 108^\circ \cos 144^\circ = 1/16$.

DIBY-42

If $A + B + C = \pi$, prove that $\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} - \sin^2 \frac{C}{2} = 1 - 2\cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$.

DIBY-43

If $A + B + C = 180^\circ$, prove that $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2\cos A \cos B \cos C$.

DIBY-44

In triangle ABC , prove that

$$\sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C) = 4\sin A \sin B \sin C.$$

DIBY-45

If $A + B + C = \pi/2$, show that

- (a) $\sin^2 A + \sin^2 B + \sin^2 C = 1 - 2\sin A \sin B \sin C$
- (b) $\cos^2 A + \cos^2 B + \cos^2 C = 2 + 2\sin A \sin B \sin C$

DIBY-46

If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then

[IIT-JEE-2009, Paper-1]

(A) $\tan^2 x = \frac{2}{3}$

(B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(C) $\tan^2 x = \frac{1}{3}$

(D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

[Ans: A,B]

DIBY-47

For $\triangle ABC$, $\sin \frac{A}{2} + \sin \frac{B}{2} + \sin \frac{C}{2} - 1 = \lambda \sin \frac{\pi-A}{4} \sin \frac{\pi-B}{4} \sin \frac{\pi-C}{4}$. Then $\lambda =$

Ans. 04

DIBY-48

The value of $\sin^3 10^\circ + \sin^3 50^\circ - \sin^3 70^\circ$ is equal to

- (A) $-\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $-\frac{3}{4}$ (D) $-\frac{3}{8}$

[Ans: D]

DIBY-49

If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$ then the value of $\cot^2 x$ is equal to _____

[Ans: 3/2]

DIBY-50

The value of $\frac{\tan(\ln 6) - \tan(\ln 2) - \tan(\ln 3)}{\tan(\ln 6) \cdot \tan(\ln 2) \cdot \tan(\ln 3)}$ is equal to _____

[Ans: 01]

DIBY-51

The value of $\cos^3\left(\frac{\pi}{8}\right) \cdot \cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right) \cdot \sin\left(\frac{3\pi}{8}\right)$ is

- (A) $\frac{1}{4}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

[JEE Mains-2020]

[Ans: B]

DIBY-52

$\tan^6\frac{\pi}{9} - 33\tan^4\frac{\pi}{9} + 27\tan^2\frac{\pi}{9}$ is equal to

[Ans: 3]

DIBY-53

If $5(\tan^2 x - \cos^2 x) = 2\cos 2x + 9$, then the value of $\cos 4x$ is

- (A) $\frac{1}{3}$ (B) $\frac{2}{9}$ (C) $-\frac{7}{9}$ (D) $-\frac{3}{5}$

[JEE Mains-2017]

[Ans: C]

DIBY-54

Find the summation of the following series

(i) $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}$

(ii) $\cos \frac{\pi}{7} + \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{5\pi}{7} + \cos \frac{6\pi}{7}$

Ans. (i) $-1/2$ (ii) 0

DIBY-55

Prove that $\sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2n-1)\theta = \frac{\sin^2 n\theta}{\sin \theta}$

DIBY-56

Sum the series : $\sqrt{1 + \cos \alpha} + \sqrt{1 + \cos 2\alpha} + \sqrt{1 + \cos 3\alpha} + \dots$ to n terms, where $0 < \alpha < \pi$

Ans. $\sqrt{2} \frac{\sin \frac{n\alpha}{4}}{\sin \frac{\alpha}{4}} \cos \left[(n+1) \frac{\alpha}{4} \right]$

DIBY-57

Find the value of

$(\cos^4 1^\circ + \cos^4 2^\circ + \cos^4 3^\circ + \dots + \cos^4 179^\circ) - (\sin^4 1^\circ + \sin^4 2^\circ + \sin^4 3^\circ + \dots + \sin^4 179^\circ).$

Ans. -1

DIBY-58

If $\frac{\sin\theta}{\cos(3\theta)} + \frac{\sin(3\theta)}{\cos(9\theta)} + \frac{\sin(9\theta)}{\cos(27\theta)} + \frac{\sin(27\theta)}{\cos(81\theta)} = \frac{\sin(k\theta)}{2\cos\theta\cos81\theta}$, then k equals to

Ans. 80

DIBY-59

$\sin\theta \cdot \sec3\theta + \sin3\theta \cdot \sec3^2\theta + \sin3^2\theta \cdot \sec3^3\theta + \dots$ upto n terms =

(A) $\frac{1}{2}[\tan3^n\theta - \tan\theta]$

(B) $\frac{1}{2}[\tan3^n\theta + \tan\theta]$

(C) $[\tan3^n\theta + \tan\theta]$

(D) $[\tan3^n\theta - \tan\theta]$

Ans. A

DIBY-60

If m and n are positive integers satisfying $1 + \cos2\theta + \cos4\theta + \cos6\theta + \cos8\theta + \cos10\theta = \frac{\cos m\theta \cdot \sin n\theta}{\sin\theta}$, then $m + n$ is equal to

Ans. 11

DIBY-61

Find the sum of series $S = \sin\theta - \sin 2\theta + \sin 3\theta - \sin 4\theta + \dots + n$ terms.

DIBY-62

The value of $P = (\sqrt{3} + \tan 1^\circ)(\sqrt{3} + \tan 2^\circ)(\sqrt{3} + \tan 3^\circ) \dots (\sqrt{3} + \tan 28^\circ)(\sqrt{3} + \tan 29^\circ)$ is

- (A) 2^{28} (B) $2^{15} + 2 - \sqrt{3}$ (C) 2^{29} (D) $2^{14} + 2 - \sqrt{3}$

Ans. (C)

*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**